PHYS 537 SOLID STATE PHYSICS

Chapter 4 Homework

Ankita Dastider

A20584591

Problem 1 – A single graphite layer

A single graphite layer is a honeycomb lattice with inequivalent sublattices A and B. Use the Bloch basis $\{|A,\mathbf{k}\rangle, |B,\mathbf{k}\rangle\}$ and include only the three nearest-neighbor A–B hoppings.

Tasks:

1. Starting from the real-space hopping sum, show that the Bloch Hamiltonian is

$$H_{1L}(k) = \begin{pmatrix} 0 & tf(k) \\ tf^*k & 0 \end{pmatrix}$$

Explain which three orbital interactions are collected into the off-diagonal matrix element.

A single graphite layer has two inequivalent carbon sites, A and B, in each primitive cell. Since the model keeps one p_z orbital on each site, the Bloch basis contains two states,

$$\{|A,k>, |B,k>\}$$

so the Hamiltonian is 2 X 2. The common on-site energy is set to zero, which gives $H_{AA} = H_{BB} = 0$. Each A site is connected to three nearest-neighbor B sites through $\delta_1, \delta_2, \delta_3$. Each hopping has amplitude t, and in a Bloch state each one also contributes a phase factor $e^{ik.\delta_j}$. Therefore,

$$H_{AB}(k) = t\sum_{j=1}^{3} e^{ik.\delta_j}$$

Using the definition,

$$f(k) = \sum_{j=1}^{3} e^{ik.\delta_j}$$

This becomes,

$$H_{AB}(k) = tf(k)$$

For the reverse B→A hopping, the phase is complex conjugated, so,

$$H_{BA}(k) = tf^*(k)$$

Thus,

$$H_{1L}(k) = \begin{pmatrix} 0 & tf(k) \\ tf^*k & 0 \end{pmatrix}$$

2. Diagonalize the matrix analytically and verify the result using numpy.linalg.eigvalsh

Starting from the Hamiltonian obtained in Task 1,

the eigenvalues are found from *det(H-EI) = 0.*

So, $det\begin{pmatrix} -E & tf(k) \\ tf^*k & -E \end{pmatrix} = 0$

$E^2 - t^2 f(k) f^*(k) = 0$

$\Rightarrow E^2 = t^2 |f(k)|^2 \qquad [\text{as } f(k) f^*(k) = |f(k)|^2]$

Therefore, the two energy bands are, $E_{\pm}(k) = \pm|t| f(k) = \pm|t f(k)|$

So, at every k, there are two energies: one negative energy band and one positive energy band. They have the same magnitude because the diagonal on-site terms are both zero.

```
f(Gamma) = (3+0j)
|f(Gamma)| = 3.0

Analytic energies (eV):
[-8.1  8.1]

Numerical energies from np.linalg.eigvalsh (eV):
[-8.1  8.1]

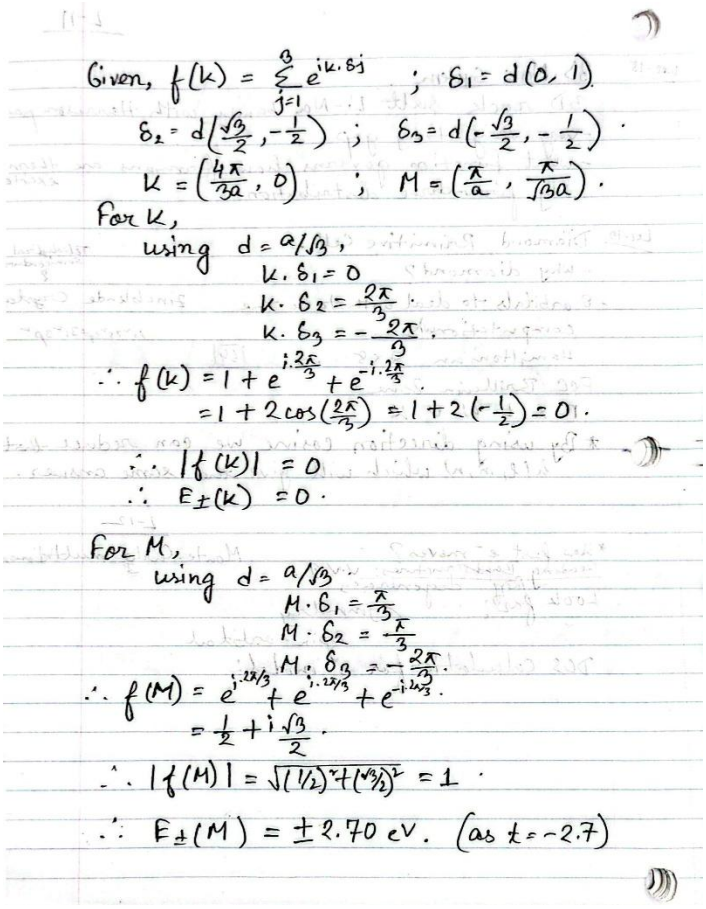
Difference:
[0. 0.]
```

At Γ, the analytic result gives (-8.10, 8.10) eV, and np.linalg.eigvalsh gives the same two eigenvalues, confirming the analytic diagonalization.

3. Use

$$\Gamma = (0,0), \qquad K = \left(\frac{4\pi}{3a}, 0\right), \qquad M = \left(\frac{\pi}{a}, \frac{\pi}{\sqrt{3}a}\right)$$

to calculate |f | and both energies at the three points.

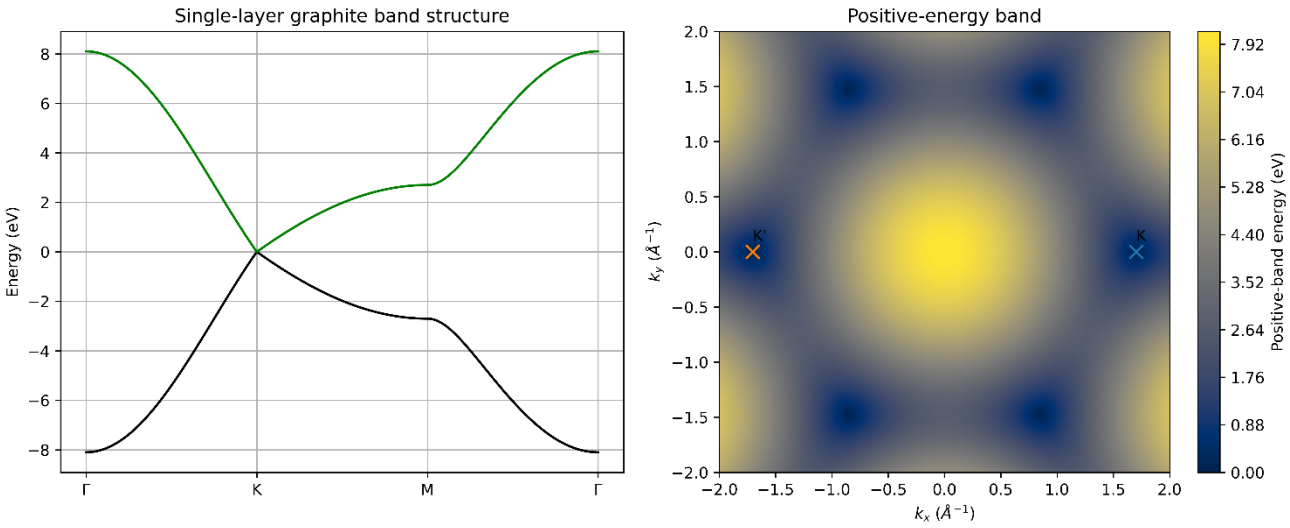


The Python calculation also confirms the expected high-symmetry values and the corresponding energies.

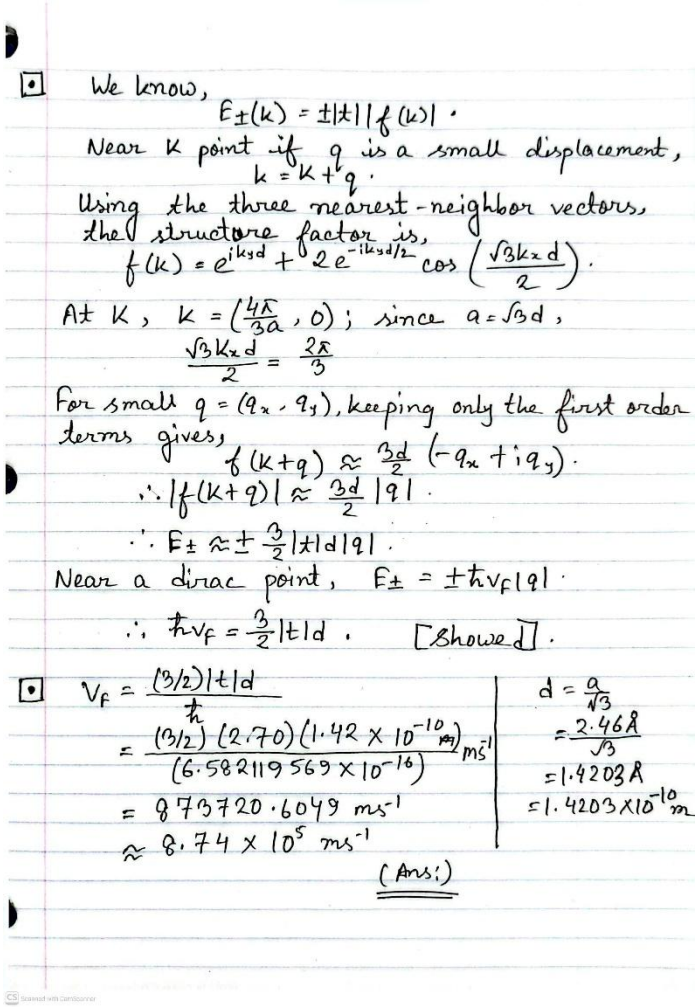
```
Point       |f(k)|  E_minus (eV)  E_plus (eV)
Gamma 3.000000e+00 -8.100000e+00 8.100000e+00
    K 4.440892e-16 -1.199041e-15 1.199041e-15
    M 1.000000e+00 -2.700000e+00 2.700000e+00
```

4. Plot both bands along Γ→ K→ M→ Γ and make a second panel showing the positive-band energy over a two-dimensional region containing K and K′. Identify the Dirac points.

The two bands along the Γ→ K→ M→ Γ path were being calculated by finding the eigenvalues of the 2 X 2 Bloch Hamiltonian at each k-point. The two bands come together at K at zero energy. In the 2D positive-energy plot, the energy also goes to zero at K and the equivalent K' point. These are the Dirac points of the single layer nearest neighbor model.



5. Expand the dispersion near K and show that $\hbar v_F = (3/2)|t|d$. Calculate v_F in ms^{-1} using $\hbar = 6.582119569 \times 10^{-16}$ eVs.




```
d = 1.4202816622064793 Angstrom
v_F = 873904.0170323352 m/s
```

6. Explain why one pz orbital on each of two sites in the primitive cell produces two bands, and why half filling places the Fermi energy at the band touching in this nearest-neighbor model.

   A primitive cell has two carbon sites, A and B, and we keep one p_z orbital from each site. Because there are two basis orbitals, the Bloch Hamiltonian is 2 X 2 and gives two eigenvalues at every k, so we get two bands. Each carbon contributes one p_z electron, giving two p_z electrons per primitive cell. With spin included, these electrons fill the lower band while the upper band remains empty. In this nearest-neighbor model, the two bands meet at the K and K' points at zero energy, so the Fermi energy lies exactly at this band touching point.

Problem 2 – The AB-stacked two-layer unit

Now place two honeycomb layers in Bernal (AB) registry. In the ordered basis $\{A_1, B_1, A_2, B_2\}$, the sites B_1 and A_2 form the vertical dimer pair. Retain t within each layer and γ_1 only between this dimer pair.

1. Construct the full 4 × 4 Bloch Hamiltonian. Label every nonzero entry by the orbital pair that produces it and verify Hermiticity.

For layer 1, the A_1-B_1 coupling is the same as in Problem 1:

$$H_{A1B1}(k) = tf(k)$$
$$H_{B1A1}(k) = tf^*(k)$$

For layer 2, the A_2-B_2 coupling is also,

$$H_{A2B2}(k) = tf(k)$$
$$H_{B2A2}(k) = tf^*(k)$$

Since B_1 sits directly above/below A_2, those two orbitals are connected by,

$$H_{B1A2} = H_{A2B1} = \gamma_1$$

All other interlayer terms are zero in this minimal model. The on-site energies are also zero, so the full Hamiltonian is,

$$H(k) = \begin{pmatrix} 0 & tf(k) & 0 & 0 \\ tf^*(k) & 0 & \gamma_1 & 0 \\ 0 & \gamma_1 & 0 & tf(k) \\ 0 & 0 & tf^*(k) & 0 \end{pmatrix}$$

with basis order $\{A_1, B_1, A_2, B_2\}$

The nonzero entries correspond to:

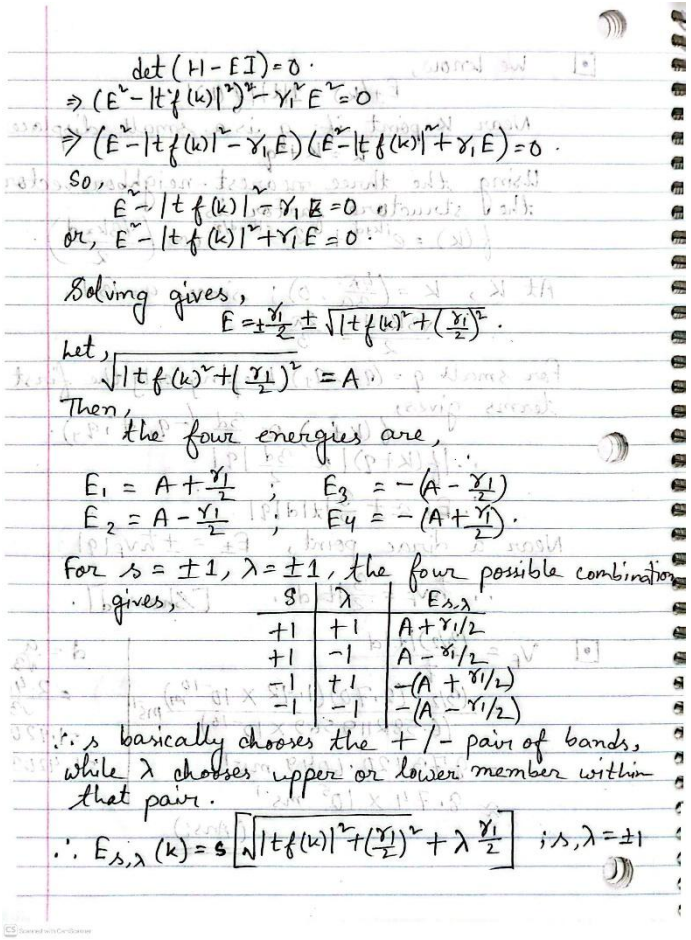
$$A_1 \leftrightarrow B_1 = tf, tf^*$$
$$B_1 \leftrightarrow A_2 = \gamma_1$$
$$A_2 \leftrightarrow 2 = tf, tf^*$$

Because every off-diagonal element has its complex conjugate in the transposed position, the matrix is Hermitian.

2. Diagonalize the matrix and show that its four energies can be written as

$$E_{s,\lambda}(k) = s\left[\sqrt{|tf(k)|^2 + \left(\frac{\gamma_1}{2}\right)^2} + \lambda\frac{\gamma_1}{2}\right], \quad s = \pm 1, \qquad \lambda = \pm 1,$$

with the four sign combinations sorted by energy.



$\det(H - EI) = 0$.

$\Rightarrow (E^2 - |tf(k)|^2)^2 - \gamma_1^2 E^2 = 0$

$\Rightarrow (E^2 - |tf(k)|^2 - \gamma_1 E)(E^2 - |tf(k)|^2 + \gamma_1 E) = 0$.

So,

$E^2 - |tf(k)|^2 - \gamma_1 E = 0$

or, $E^2 - |tf(k)|^2 + \gamma_1 E = 0$.

Solving gives,

$E = \pm\frac{\gamma_1}{2} \pm \sqrt{|tf(k)|^2 + \left(\frac{\gamma_1}{2}\right)^2}$.

Let,

$\sqrt{|tf(k)|^2 + \left(\frac{\gamma_1}{2}\right)^2} = A$.

Then, the four energies are,

$E_1 = A + \frac{\gamma_1}{2}$; $E_3 = -\left(A - \frac{\gamma_1}{2}\right)$

$E_2 = A - \frac{\gamma_1}{2}$; $E_4 = -\left(A + \frac{\gamma_1}{2}\right)$.

For $s = \pm 1$, $\lambda = \pm 1$, the four possible combinations gives,

s	λ	$E_{s,\lambda}$
+1	+1	$A + \gamma_1/2$
+1	−1	$A - \gamma_1/2$
−1	+1	$-(A + \gamma_1/2)$
−1	−1	$-(A - \gamma_1/2)$

$\therefore s$ basically chooses the +/− pair of bands, while λ chooses upper or lower member within that pair.

$\therefore E_{s,\lambda}(k) = s\left[\sqrt{|tf(k)|^2 + \left(\frac{\gamma_1}{2}\right)^2} + \lambda\frac{\gamma_1}{2}\right]$; $s, \lambda = \pm 1$

The equation above gives the four energy bands from the analytic calculation. We can also compare this formula with np.linalg.eigvalsh to calculate the energies numerically. The analytic and numerical values should agree to numerical precision.

```
Analytic energies:
[-8.29734688 -7.90734688  7.90734688  8.29734688]

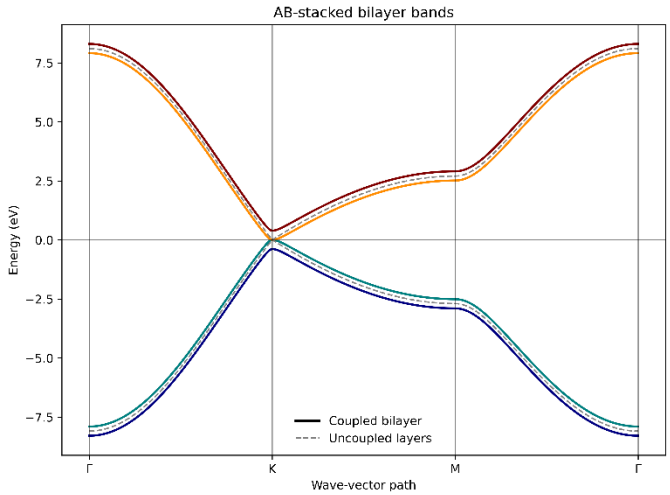
Numerical energies:
[-8.29734688 -7.90734688  7.90734688  8.29734688]

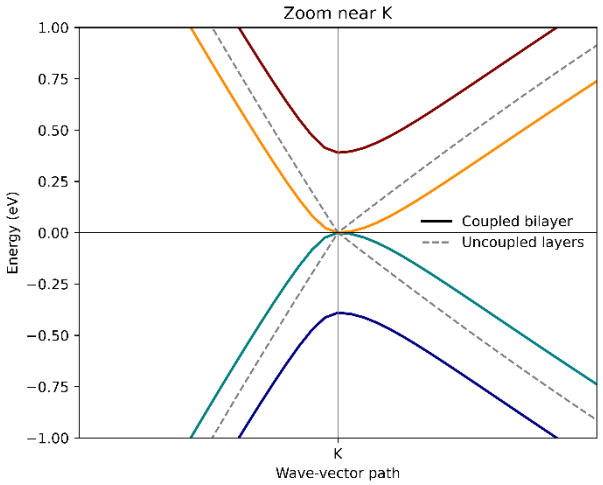
Difference:
[ 0.00000000e+00 -8.88178420e-16  1.77635684e-15  1.77635684e-15]
```

The numerical values match the analytic result, so the diagonalization is consistent.

3. Plot all four bands along Γ→ K→ M→ Γ. Compare them with two uncoupled graphite layers.

For the coupled bilayer, $\gamma_1 = 0.390$ eV represents the hopping between the B_1 and A_2 sites of the two layers. To see what changes because of this interaction, the same band structure can be calculated again with $\gamma_1 = 0$ eV. This removes the vertical coupling, so the two layers behave independently and give two identical copies of the single layer bands. Comparing the two cases makes the effect of the interlayer coupling clear, especially near K, where some of the bands split once γ_1 is included.





In the uncoupled case, there are still four eigenvalues, but they form two identical pairs because the two layers have the same single layer bands. Since each pair lies on top of each other, only two dashed curves are visible. When γ_1 is included, this degeneracy is partly removed and the four coupled bands become separated. The difference is easiest to see near K, where two states move away from zero energy while the other two remain at zero.

4. At K, report the four eigenvalues and the weights $|cA_1|^2$, $|cB_1|^2$, $|cA_2|^2$, and $|cB_2|^2$ for every eigenstate. Display these weights as a labeled heat map.

At K=0,

$$f(K) = 0$$

$$\therefore H(K) = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & \gamma_1 & 0 \\ 0 & \gamma_1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

Only B_1 and A_2 are directly coupled at K, while A_1 and B_2 are not.

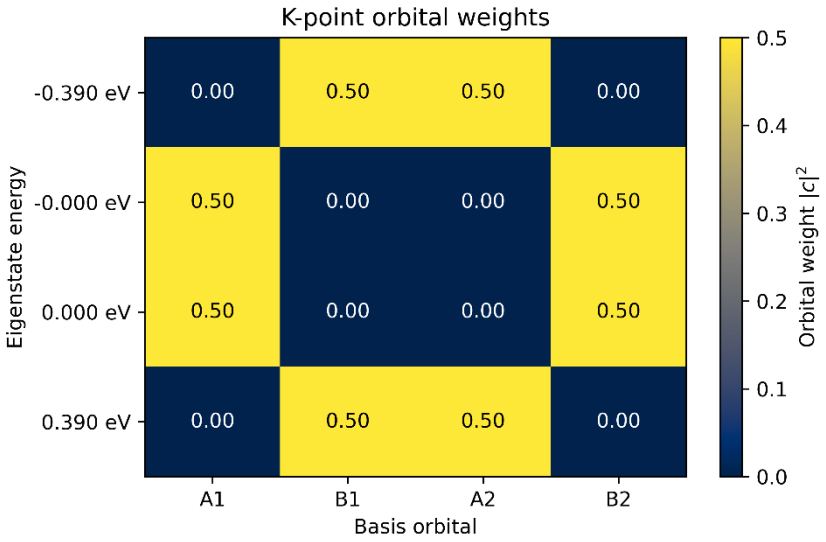
Therefore, the four eigenvalues are, $+\gamma_1, 0, 0, -\gamma_1 = +0.390, 0, 0, -0.390$ eV.

The two split states are equal mixtures of B_1 and A_2. The only difference is their relative sign. One is the symmetric combination, and the other is the antisymmetric combination.

$$|c_{B_1}|^2 = |c_{A_2}|^2 = \frac{1}{2}$$

The two zero-energy states come from the non-dimer sites A_1 and B_2.

Energy (eV)	A1	B1	A2	B2
-3.900000e-01	4.726164e-30	5.000000e-01	5.000000e-01	4.726164e-30
-3.686408e-30	5.000000e-01	4.726164e-30	4.726164e-30	5.000000e-01
3.686408e-30	5.000000e-01	4.726164e-30	4.726164e-30	5.000000e-01
3.900000e-01	4.726164e-30	5.000000e-01	5.000000e-01	4.726164e-30



The two states at 0 eV have the same energy, so the eigensolver can choose different combinations of A_1 and B_2 rather than making one state purely A_1 and the other purely B_2. In this result, both zero-energy states have equal weight on A_1 and B_2. The main point is that these states come only from the non-dimer sites A_1 and B_2, while the states at $\pm\gamma_1$ come from the coupled B_1 and A_2 sites.

5. Use the K-point eigenvectors to distinguish the dimer sites from the non-dimer sites. Explain why two states remain at zero energy while two are split by the vertical coupling.

   B_1 and A_2 are called the dimer sites because they are directly connected to each other through the vertical hopping term γ_1. In contrast, A_1 and B_2 do not have this direct interlayer connection, so they are the non-dimer sites.
   From the K-point orbital weights in Task 4, the two split states mainly come from the dimer pair B_1 and A_2, while the zero-energy states come from the non-dimer sites A_1 and B_2. At K, f(K)=0, so the in-plane hopping terms disappear. The only coupling left is the vertical γ_1 term between B_1 and A_2, which splits their two combinations to energies $-\gamma_1$ and $+\gamma_1$. Since A_1 and B_2 are not connected by this vertical hopping, their states are not shifted and remain at zero energy.

6. Explain how the change from two orbitals to four orbitals in the repeat unit changes the number of bands without changing the basic Bloch-block construction.

   For a single graphite layer, there are two p_z orbitals in the primitive cell, one on A and one on B. That gives a 2 X 2 Bloch Hamiltonian, so there are two energy bands. In the AB-stacked bilayer, the repeat unit contains four p_z orbitals, $\{A_1,B_1,A_2,B_2\}$, so the Hamiltonian becomes 4 X 4 and gives four bands. The main idea of the Bloch construction does not change when the basis gets larger. Each matrix element still comes from either an on-site term or a hopping interaction, with the appropriate Bloch phase included for hopping between cells. The extra orbitals simply add more rows and columns to the Hamiltonian and introduce additional couplings, such as the vertical γ_1 hopping between B_1 and A_2. So, the number of bands increases because there are more basis orbitals in the repeat unit, while the method used to build the Bloch Hamiltonian stays the same.

Problem 3 – Three-dimensional Bernal graphite

Restore periodicity along z. A B_1 orbital couples to equivalent A_2 orbitals in the adjacent two-layer repeats above and below. Their two Bloch phases combine into

$$g(k_z) = \gamma_1 e^{ik_z c/2} + \gamma_1 e^{-ik_z c/2} = 2\gamma_1 \cos\left(\frac{k_z c}{2}\right). \tag{7}$$

Use the four-orbital basis $\{A_1, B_1, A_2, B_2\}$ and replace the bilayer dimer element by $g(k_z)$.

Tasks:

1. Write and construct the complete 4 ×4 bulk Hamiltonian H(k) for arbitrary (kx,ky,kz). Explain explicitly why two physically distinct vertical hoppings appear in one matrix element.

For the 3D Bernal graphite model, the same four-orbital basis is used,

$$\{A_1, B_1, A_2, B_2\}$$

The in-plane hopping is unchanged from the bilayer case and is still described by tf(k). The main difference is in the vertical $B_1 - A_2$ coupling. In bulk graphite, a B_1 orbital couples to equivalent A_2 orbitals in the neighboring repeats above and below. These two hoppings contribute the Bloch phase factors $e^{ik_z c/2}$ and $e^{-ik_z c/2}$ so their contributions add:

$$g(k_z) = \gamma_1 e^{ik_z c/2} + \gamma_1 e^{-ik_z c/2} = 2\gamma_1 \cos\left(\frac{k_z c}{2}\right)$$

Therefore, the bilayer coupling γ_1 is replaced by $g(k_z)$, and the bulk Hamiltonian becomes

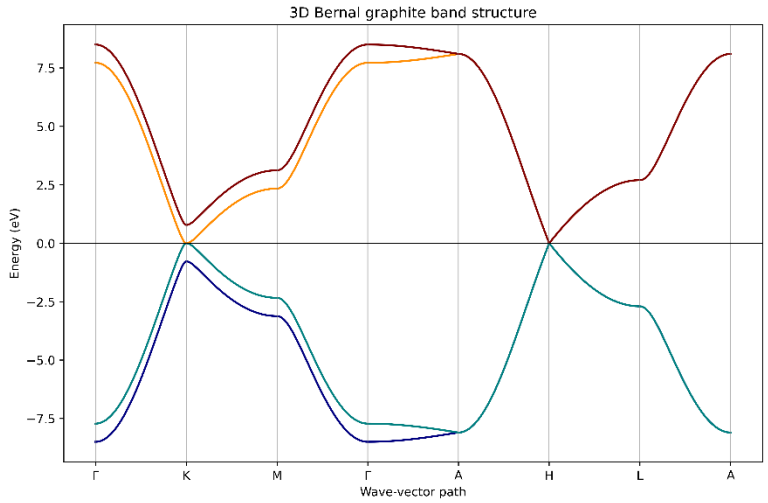
$$H(\boldsymbol{k}) = \begin{pmatrix} 0 & tf(\boldsymbol{k}) & 0 & 0 \\ tf^*(\boldsymbol{k}) & 0 & g(k_z) & 0 \\ 0 & g(k_z) & 0 & tf(\boldsymbol{k}) \\ 0 & 0 & tf^*(\boldsymbol{k}) & 0 \end{pmatrix}$$

in the basis $\{A_1, B_1, A_2, B_2\}$ and $\mathbf{k}=(k_x, k_y, k_z)$. The two vertical hoppings appear in the same matrix element because both represent coupling between the same pair of basis orbitals, B_1 and A_2. The only difference is that one coupling is to the equivalent site above and the other is to the equivalent site below. Their different positions give opposite Bloch phases, and after adding them together they produce the single k_z dependent term $g(k_z)$. This is what introduces the third-direction periodicity into the graphite Hamiltonian.

```
Hamiltonian:
[[ 0.        +0.j         -6.76945899-0.18338574j  0.        +0.j
   0.        +0.j        ]
 [-6.76945899+0.18338574j  0.        +0.j          0.6113809 +0.j
   0.        +0.j        ]
 [ 0.        +0.j          0.6113809 +0.j          0.        +0.j
  -6.76945899-0.18338574j]
 [ 0.        +0.j          0.        +0.j         -6.76945899+0.18338574j
   0.        +0.j        ]]

Maximum |H - H†| = 0.0
```

2. Plot all four bands along Γ →K→M→Γ→A→H→L→A, where *A = (Γ,π/c), H = (K,π/c),* and *L = (M,π/c).*



The four energy bands were calculated along the high-symmetry path Γ →K→M→Γ→A→H→L→A using the 4 X 4 bulk Hamiltonian from Task 1. The first part of the path, Γ →K→M→Γ is in the $k_z = 0$ plane, while A→H→L→A is in the $k_z = \pi/c$ plane. In the second part, the bands appear as two overlapping pairs because $g(k_z) = 0$ at $k_z = \pi/c$ so the two layers become degenerate in this minimal model. At H, all four bands meet at zero energy.

3. Verify the limiting forms at K and H. Explain why the two layers are maximally coupled at $k_z = 0$ but decouple at $k_z = \pi/c$.

At K,

$$k_z = 0, \qquad f(K) = 0, \qquad g(0) = 2\gamma_1 \cos(0) = 2\gamma_1$$

$$\therefore H(K) = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 2\gamma_1 & 0 \\ 0 & 2\gamma_1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

Only B_1 and A_2 remained coupled at K.

Therefore, the four eigenvalues are, $+ 2\gamma_1, 0, 0, - 2\gamma_1 = + 0.780, 0, 0, - 0.780$ eV.

At H,

$$k_z = \frac{\pi}{c}, \qquad f(H) = 0, \qquad g\left(\frac{\pi}{c}\right) = 2\gamma_1 \cos\left(\frac{\pi}{2}\right) = 0$$

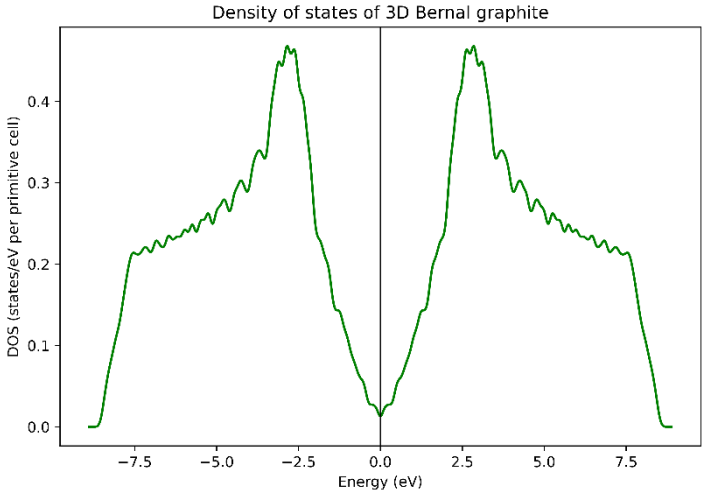
$$\therefore H(H) = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

Therefore, all the four eigenvalues are 0, 0, 0, 0.

The difference between K and H comes from how the two vertical hopping terms combine. At $k_z = 0$, the two contributions have the same phase, so they reinforce each other and give the largest coupling, $g = 2\gamma_1$. At $k_z = \frac{\pi}{c}$, they have opposite phases and cancel each other. This makes g = 0, so the two layers are effectively decoupled in this simple model.

```
K-point energies = [-7.80000000e-01 -1.84320385e-30  1.84320385e-30  7.80000000e-01]
H-point energies = [-1.22315926e-15 -1.17539804e-15  1.17539804e-15  1.22315926e-15]
```

4. Calculate a full-zone DOS on a shifted uniform mesh spanning one primitive reciprocal cell. State the mesh, broadening, and DOS normalization. Verify that the DOS integrates to four states per primitive cell when spin is not counted.



The full-zone DOS was calculated using a shifted 30 X 30 X 30 uniform mesh covering one primitive reciprocal cell. The half-grid shift places each k-point at the center of an equal-volume reciprocal-space cell and avoids sampling equivalent points on opposite boundaries. At every k-point, the four eigenvalues of the bulk Hamiltonian were calculated using np.linalg.eigvalsh. The discrete states were broadened using normalized Gaussians with σ = 0.08 eV. Since spin is not included, each sampled eigenstate was given a weight $1/N_k$. The numerical integration of the DOS gives approximately 4 states per primitive cell, which agrees with the four p_z orbitals in the basis and confirms the DOS normalization.

```
Gaussian broadening = 0.08 eV
Integrated DOS = 3.9999999997148135
```

*ChatGPT is being used to understand this task.

5. Estimate a vertical pz–pz sigma hopping with Harrison's rule

$$V_{pp\sigma} = 2.22\frac{\hbar^2}{m_e d^2}\,, \qquad \frac{\hbar^2}{m_e} = 7.62\ eV\text{Å}^2\,, \qquad d_\perp = 3.35\ \text{Å}$$

Compare it with γ_1 = 0.390 eV and explain why the empirical graphite value is more appropriate for this weak interlayer bond.

Here,

$$V_{pp\sigma} = 2.22\frac{\hbar^2}{m_e d^2} = 2.22\frac{7.62}{(3.35)^2} \approx 1.51\ eV$$

But the empirical value that was used is γ_1 = 0.390 eV.

The Harrison estimate is therefore $\frac{1.51}{0.390} \approx 3.87$ times larger.

The empirical value is more suitable here because it is meant to represent the actual interaction between neighboring graphene layers. That interaction is weak compared with the strong bonding within each layer. Harrison's rule only gives a rough estimate based mainly on the orbital type and the distance between the atoms, so it does not fully capture the specific environment of graphite. Because of that, it gives a much larger hopping value. For this model, γ_1 is therefore a better choice because it describes the real interlayer coupling in graphite more realistically.

6. State what this minimal model predicts at the Fermi energy and name at least two omitted hopping terms or physical effects needed for a quantitatively accurate graphite Fermi surface.

In this model, the Fermi energy is at where the valence and conduction bands touch. Along the K - H direction, two bands stay at zero energy, so there is no band gap in this simple model. However, only the nearest-neighbor in-plane hopping t and the main interlayer hopping γ_1 are included here. In real graphite, other longer range hopping interactions are also present, and these can change the bands near the Fermi energy. Because of that, this model is useful for showing the basic band structure, but it is not enough to describe the real graphite Fermi surface accurately.

For a more accurate graphite Fermi surface, longer-range in-plane hopping and additional interlayer hopping beyond would need to be included. These extra interactions can shift and reshape the bands close to the Fermi energy.